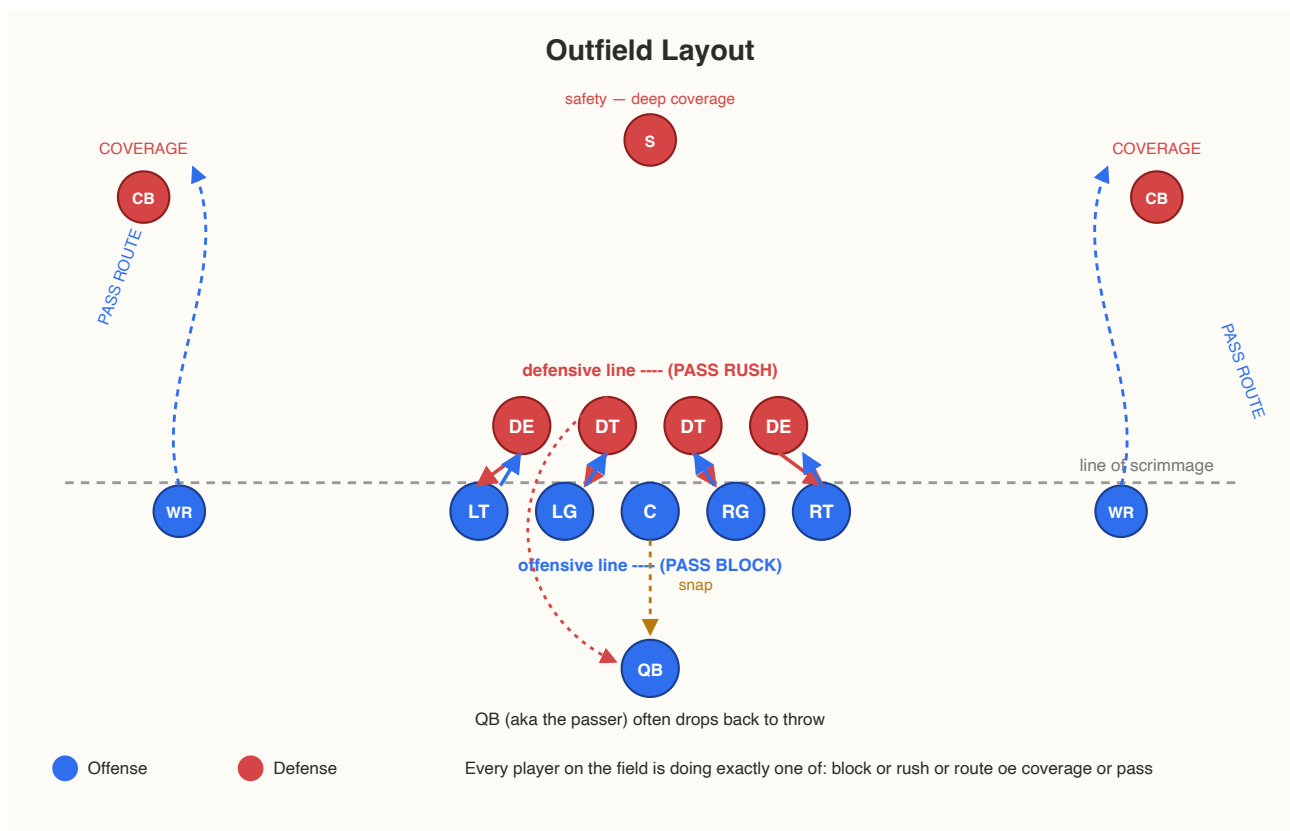


Basic how it works:

Things to keep in mind:

- The game consists of discrete plays.
- Two teams with 11 players each line up facing each other then
 - One team has the ball (**offense**)
 - The other team tries to take it away or stop its progress (**defense**)
- A play starts with the **snap** — the center or an offensive player passes the ball backward between his legs to the quarterback and ends a few seconds later when the play is whistled dead (tackle, ball goes out of bounds, pass falls incomplete, score).
- The field is 100 yards of playing field plus a 10-yard **end zone** at each end (120 yards total), and about 53 yards wide. The offense has **4 downs** (attempts) to advance the ball 10 yards; if they succeed, they get a fresh set of 4 downs.
- On any given play, the offense either **runs** the ball (hands it to a runner) or **passes** it (the QB throws it downfield)



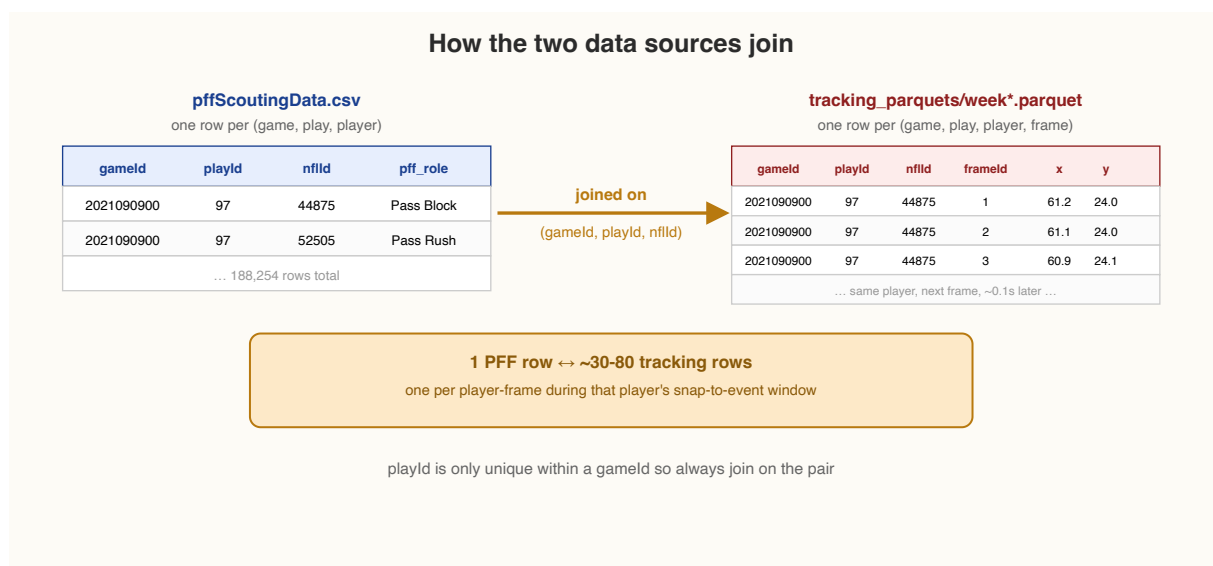
Dataset review.

- Dataset was extracted from <https://www.kaggle.com/datasets/dmay01/usefuldata>

It only has the following files:

- pffScoutingData.csv
- team_information.csv
- tracking_parquets/week1.parquet ... week8.parquet

How the two files relate to each other



pffScoutingData.csv and the tracking parquet files describe the *same plays* from two different perspectives:

- pffScoutingData.csv is **one row per (game, play, player)** — a human's judgment of what that player did and how well it worked. It has no sense of time within the play.
- The tracking parquets are **one row per (game, play, player, 1/10th-of-a-second frame)** — the raw (x, y) position, speed, acceleration, etc. of every player, 10 times a second, for the whole play.

Furthermore:

- The dataset only contains **pass plays**.
- Every play in this data follows the same basic shape:
 1. **Snap** — the center hikes the ball to the QB.
 2. **Pass protection** — while the QB looks to throw, 5 offensive linemen try to stop defenders from reaching him. This is called **pass blocking**.
 3. **Pass rush** — the defenders trying to reach and tackle the QB before he throws are **pass rushing**.
 4. **Routes and coverage** — meanwhile, offensive receivers run prescribed paths (**routes**) to get open for a catch, while defensive backs try to stay attached to them (**coverage**).
 5. **Resolution** — the QB throws the ball (completed, incomplete, or intercepted), or a pass rusher reaches him first (sack), or he scrambles (runs with it himself).
- Every player on the field is doing exactly one of those things on a pass play either pass blocking, pass rushing, running a route, or playing coverage (plus the QB, who is just technically “passing”).
- That’s the taxonomy which was used to label every player on every play in `pffScoutingData.csv`

Positions found in dataset

1. **Offensive line**
2. **Left Tackle (LT),**
3. **Left Guard (LG),**
4. **Center (C),**
5. **Right Guard (RG),**
6. **Right Tackle (RT).**
 - Tight ends (TE) and running backs (HB/FB) sometimes help block too
7. **Defensive pass rushers:** defensive ends or tackles (**DE, DT, DLT, DRT, LEO, REO, NT, and the list goes on** and sometimes linebackers rushing instead of dropping into coverage (a **blitz**).
- **Everyone else:**
 8. **wide receivers (WR)**
 9. **tight ends (TE)** run routes
 10. **cornerbacks (CB)**
 11. **safeties (S/FS/SS)** play coverage
 12. **the quarterback (QB)** is the passer.

How I used dataset files:

I used the `pffScoutingData.csv` file to get the play-level judgment (“this blocker allowed a hurry”) and reconstructing *what actually happened, physically, in those 2-3 seconds* from the raw tracking data (how close did the rusher get? how fast were they moving? did the gap close over time?).

How to classify plays or in other words “success” and “failure”

- **Pressure** — an umbrella term for the pass rush disrupting the QB before he can comfortably throw. It comes in three flavors, worst to mildest:
 - **Sack** — a pass rusher tackles the QB *behind the line of scrimmage* before he throws. Worst outcome for the offense: loss of yardage and the down is used up.
 - **Hit** — the QB is hit (often right as or just after he throws) but isn’t sacked.
 - **Hurry** — the QB is forced to rush, step up, or alter his throw because a rusher got close, without being hit or sacked. The mildest form of pressure, but still a defensive win.
- From the **blocker’s side**, letting any of these three happen is a failure: the data has `pff_hitAllowed` / `pff_hurryAllowed` / `pff_sackAllowed` flags on blockers.
- From the **rusher’s side**, causing any of these three is a success: the data has `pff_hit` / `pff_hurry` / `pff_sack` flags on rushers.
- **“Beaten”** (`pff_beatenByDefender`) is considered when a blocker lost his one-on-one matchup — a slightly broader/softer notion than the three flags above, since a blocker can be beaten even if the QB got the ball out before it turned into a hit/hurry/sack.

Useful links

- [Dictionary of terminologies and columns](#)
- [Use Cases](#)
- [Tracking Data System](#)